Here the critical path is 1–2–4–6

The project completion time is 25 weeks

Exercise 10.2

1. Draw the network for the project whose activities with their relationships are given below:

   Activities A,D,E can start simultaneously; B,C>A; G,F>D,C; H>E,F.

2. Draw the event oriented network for the following data:

Events	1	2	3	4	5	6	7
Immediate Predecessors	–	1	1	2,3	3	4,5	5,6

3. Construct the network for the projects consisting of various activities and their precedence relationships are as given below:

   A,B,C can start simultaneously A<F, E; B<D, C; E, D<G

4. Construct the network for each the projects consisting of various activities and their precedence relationships are as given below:

Activity	A	B	C	D	E	F	G	H	I	J	K
Immediate Predecessors	–	–	–	A	B	B	C	D	E	H,I	F,G

5. Construct the network for the project whose activities are given below.

Activity	0-1	1-2	1-3	2-4	2-5	3-4	3-6	4-7	5-7	6-7
Duration (in week)	3	8	12	6	3	3	8	5	3	8

   Calculate the earliest start time, earliest finish time, latest start time and latest finish time of each activity. Determine the critical path and the project completion time.

6. A project schedule has the following characteristics

Activity	1-2	1-3	2-4	3-4	3-5	4-9	5-6	5-7	6-8	7-8	8-10	9-10
Time	4	1	1	1	6	5	4	8	1	2	5	7

   Construct the network and calculate the earliest start time, earliest finish time, latest start time and latest finish time of each activity and determine the Critical path of the project and duration to complete the project.

7. Draw the network and calculate the earliest start time, earliest finish time, latest start time and latest finish time of each activity and determine the Critical path of the project and duration to complete the project.

Jobs	1-2	1-3	2-4	3-4	3-5	4-5	4-6	5-6
Duration	6	5	10	3	4	6	2	9

8. The following table gives the activities of a project and their duration in days

Activity	1-2	1-3	2-3	2-4	3-4	3-5	4-5
Duration	5	8	6	7	5	4	8

   Construct the network and calculate the earliest start time, earliest finish time, latest start time and latest finish time of each activity and determine the Critical path of the project and duration to complete the project.

9. A Project has the following time schedule

Activity	1-2	1-6	2-3	2-4	3-5	4-5	6-7	5-8	7-8
Duration (in days)	7	6	14	5	11	7	11	4	18

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